Module 7

Option Pricing – Black-Scholes-Merton Equation

October 8 & 13, 2025

Stochastic Process for Stock Return

Concepts in stochastic process & stochastic calculus:

- Markovian property of a stochastic process:

  Review of Markov chain:

  Let $x_0, x_1, x_2, ..., x_n$ be a sequence of random variables, it is Markov chain if

  $P(x_{n+1} \mid x_n, x_{n-1}, ..., x_1, x_0) = P(x_{n+1} \mid x_n)$

  → to forecast future only present information matters

- Wiener process (Brownian motion, random walk) W_t:

  For a small increment in time δt,

  $W_t \to W_{t+\delta t} = W_t + \delta W_t$, $\delta W_t = \varepsilon\sqrt{\delta t}$, $\varepsilon \sim N(0,1)$

  $E(\delta W_t) = 0$, $E(\delta W_t^2) - E(\delta W_t)^2 = \delta t$, $E(\delta W_t \delta W_{t'}) = 0$ $(t \neq t')$

  $N(0,1)$: standard normal distribution, $E(.)$: expectation

  Now let $\delta t = \dfrac{t}{n}$

  $W_t - W_0 = W_t - W_{(n-1)\delta t} + W_{(n-1)\delta t} - ... - W_{2\delta t} + W_{2\delta t} - W_{\delta t} + W_{\delta t} - W_0$

  $= \sum_{i=0}^{n-1} \varepsilon_i \sqrt{\delta t}$ $\quad (E(\varepsilon_i \varepsilon_j) = 0)$

  $\sim N(0,t)$ $\quad (n \to \infty$, central limit theorem)

  Let $W_0 = 0$ → $E(W_t) = 0$, $E(W_t^2) = t$, $W_t \sim N(0,t)$

- Generalized Wiener process (random walk with drift)

  $\delta x_t = a\delta t + b\delta W_t$

- Ito process:

  $\delta x_t = a(x_t, t)\delta t + b(x_t, t)\delta W_t$

  In continuous time:

  $dx_t = a(x_t, t)dt + b(x_t, t)dW_t$

  To simplify notation we shall drop the subscript t in x if no confusion:

  $dx = a(x, t)dt + b(x, t)dW_t$

  Example:

  Modeling stock price movement using Ito process

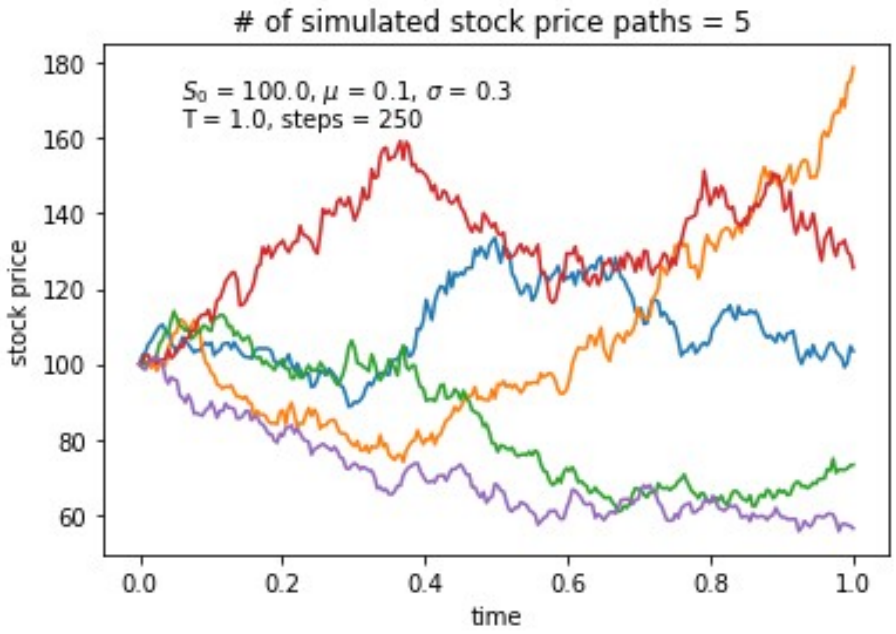
  $\frac{dS}{S} = \mu dt + \sigma dW_t$

  If $\sigma = 0$:

  $\frac{dS}{S} = \mu dt \rightarrow \int \frac{1}{S} dS = \int d(\ln S) = \int \mu dt$

  $\rightarrow \ln\left(\frac{S_t}{S_0}\right) = \int_0^t \mu ds \rightarrow S_t = S_0 e^{\mu t}$ (μ constant)

  σdW : randomness in stock return



- Ito Integral:

  Divide $(0,t)$ into $0 = t_0 < t_1 < ... < t_{n-1} < t_n = t$, $t_i - t_{i-1} = \delta t = \frac{t}{n}$

  For a "regular" function f:

$$\int_0^t f(W_s, s) dW_s = \lim_{n\to\infty} \sum_{j=1}^{n} f(W_{j-1}, t_{j-1}) \left(W_j - W_{j-1}\right)$$

  Different from Riemann-Stieltjes integral, the selection of point in $[t_{j-1}, t_j)$ at which we compute f matters

  Example:

$$\int_0^t W_s dW_s = ? \quad (W_0 = 0)$$

$$\sum_{j=1}^{n} W_{j-1}\left(W_j - W_{j-1}\right) = \sum_{j=1}^{n} W_{j-1}W_j - \sum_{j=1}^{n} W_{j-1}^2$$

$$= \sum_{j=1}^{n} W_{j-1}W_j - \frac{1}{2}\left\{\sum_{j=1}^{n} W_{j-1}^2 + \sum_{j=1}^{n} W_j^2 - W_t^2\right\} \quad (W_0 = 0, W_n = W_t)$$

$$= \frac{1}{2}W_t^2 - \frac{1}{2}\sum_{j=1}^{n}\left(W_j - W_{j-1}\right)^2 = \frac{1}{2}W_t^2 - \frac{1}{2}\sum_{j=1}^{n}\left(\varepsilon_{j-1}\sqrt{\delta t}\right)^2$$

$$= \frac{1}{2}W_t^2 - \frac{t}{2n}\sum_{j=0}^{n-1}\varepsilon_j^2$$

$$= \frac{1}{2}W_t^2 - \frac{1}{2}t \quad (n \to \infty, \text{ strong law of large number})$$

→ $\int_0^t W_s dW_s = \frac{1}{2}W_t^2 - \frac{1}{2}t$

One can verify that

$$\lim_{n\to\infty} \sum_{j=1}^{n} W_j\left(W_j - W_{j-1}\right) = W_t^2 - \lim_{n\to\infty}\left\{\sum_{j=1}^{n} W_{j-1}W_j - \sum_{j=1}^{n} W_{j-1}^2\right\} = \frac{1}{2}W_t^2 + \frac{1}{2}t$$

→ where do we take f value matters!

e.g., $\lim_{n\to\infty}\sum_{j=1}^{n}\left(\frac{W_j+W_{j-1}}{2}\right)\left(W_j-W_{j-1}\right)$ → "Stratonovich integral"

Luckily we do not need to compute every Ito integral from the above definition thanks to Ito's lemma.

- Ito's Lemma:

  Assume x follows an Ito process,

  $dx = a(x,t)dt + b(x,t)dW_t$

  then for $G(x,t)$,

  $$dG(x,t) = \left(\frac{\partial G}{\partial t} + \frac{1}{2}b^2\frac{\partial^2 G}{\partial x^2}\right)dt + \frac{\partial G}{\partial x}dx$$

  $$= \left(\frac{\partial G}{\partial t} + a\frac{\partial G}{\partial x} + \frac{1}{2}b^2\frac{\partial^2 G}{\partial x^2}\right)dt + b\frac{\partial G}{\partial x}dW_t$$

  $\frac{1}{2}b^2\frac{\partial^2 G}{\partial x^2}dt$: extra term in Ito calculus

  A heuristic "derivation":

  From calculus:

  $$\delta G(x,t) = \frac{\partial G}{\partial t}\delta t + \frac{\partial G}{\partial x}\delta x + \frac{1}{2}\frac{\partial^2 G}{\partial t^2}(\delta t)^2 + \frac{1}{2}\frac{\partial^2 G}{\partial x^2}(\delta x)^2 + \frac{1}{2}\frac{\partial^2 G}{\partial t\partial x}\delta t\delta x + ...$$

  Here $\delta x = a(x,t)\delta t + b(x,t)\delta W_t$, $\delta W_t = \varepsilon\sqrt{\delta t}$. To the leading order of δt

  $$\delta G(x,t) = \frac{\partial G}{\partial t}\delta t + \frac{\partial G}{\partial x}(a\delta t + b\delta W_t) + \frac{1}{2}\frac{\partial^2 G}{\partial x^2}(b\delta W_t)^2$$

  δW_t^2 is a random variable with $E(\delta W_t^2) = \delta t$.

  How random is δW_t^2 ?

  Variance of $\delta W_t^2 = E(\delta W_t^4) - E(\delta W_t^2)^2 = 3(\delta t)^2 - (\delta t)^2 = 2(\delta t)^2$

  → not very random at the scale of δt → $\delta W_t^2 \approx \delta t$

  → $\delta G(x,t) = \left(\frac{\partial G}{\partial t} + a\frac{\partial G}{\partial x} + \frac{1}{2}b^2\frac{\partial^2 G}{\partial x^2}\right)\delta t + b\frac{\partial G}{\partial x}\delta W_t$

Examples:

- $\int_0^t W_s dW_s = ?$

  $G = W_t^2 \rightarrow d\left(W_t^2\right) = 2W_t dW_t + dt$

  $\rightarrow \int_0^t W_s dW_s = \frac{1}{2}\left\{\int_0^t d\left(W_t^2\right) - \int_0^t dt\right\} = \frac{W_t^2}{2} - \frac{t}{2},\ (W_0 = 0)$

  (extra term comparing to Riemann-Stieltjes integral)

- $\int_0^t s dW_s = ?$

  $G = tW \rightarrow d\left(tW_t\right) = tdW_t + W_t dt \rightarrow \int_0^t s dW_s = tW_t - \int_0^t W_s ds$

- Lognormal property of the stock process:

  For stock process S (assume div = 0, and for simplicity, μ, σ constant):

  $dS = \mu S dt + \sigma S dW_t$

  $\rightarrow dG(S,t) = \left(\frac{\partial G}{\partial t} + \mu S \frac{\partial G}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 G}{\partial S^2}\right)dt + \sigma S \frac{\partial G}{\partial S} dW_t$

  Now

  $\frac{dS}{S} = \mu dt + \sigma dW_t \rightarrow S(t,W) = ?$

  Without randomness,

  $\frac{dS}{S} = \mu dt \rightarrow S_t = S_0 e^{\mu t}$

  With randomness, first guess:

  $S_t = S_0 e^{\mu t + \sigma W_t}$ (?)

  $\rightarrow dS = \left(\mu + \frac{\sigma^2}{2}\right)S dt + \sigma S dW_t$ → extra term from Ito

  Now try

$S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t}$ → works!

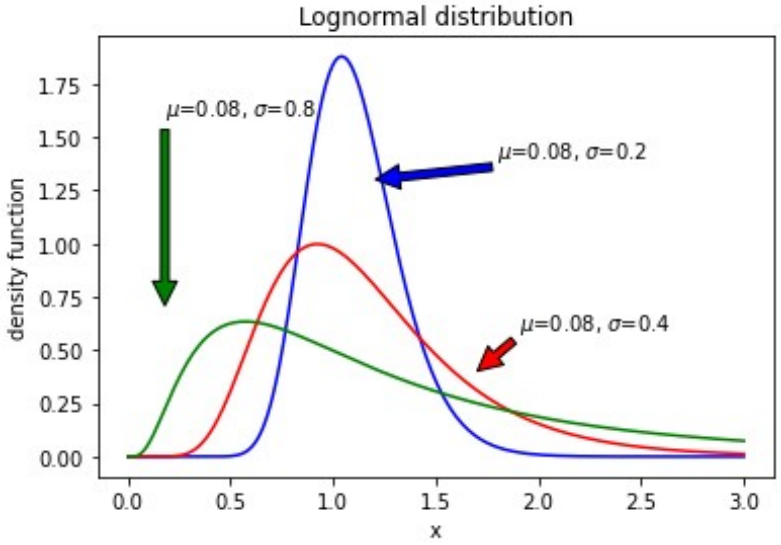
→ $\ln S_t = \ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t$ → $\ln S_t \sim N\left[\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right]$

$$P(\ln S_t \le \ln Y) = \int_{-\infty}^{\ln Y} \frac{1}{\sigma\sqrt{2\pi t}} e^{-\frac{\left(u - \ln S_0 - \left(\mu - \frac{\sigma^2}{2}\right)t\right)^2}{2\sigma^2 t}} du = \int_0^{Y} \frac{1}{\sigma x\sqrt{2\pi t}} e^{-\frac{\left(\ln x - \ln S_0 - \left(\mu - \frac{\sigma^2}{2}\right)t\right)^2}{2\sigma^2 t}} dx$$

→ $$P(S_t \le Y) = \int_0^{Y} \frac{1}{\sigma x\sqrt{2\pi t}} e^{-\frac{\left(\ln x - \ln S_0 - \left(\mu - \frac{\sigma^2}{2}\right)t\right)^2}{2\sigma^2 t}} dx$$

Probability density function for lognormal distribution with parameters (μ, σ):

$$g(x;\mu,\sigma) = \frac{1}{\sigma x\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$



Consequences of the lognormal property:

- $E(S_t) = S_0 e^{\mu t}$, $E(S_t^2) - E(S_t)^2 = S_0^2 e^{2\mu t}(e^{\sigma^2 t} - 1)$ (exercise)
- Let x = continuous compounded rate of return for stock S: $S_t = S_0 e^{xt}$

$$x = \frac{1}{t}\ln\left(\frac{S_t}{S_0}\right) \sim N\left[\left(\mu - \frac{\sigma^2}{2}\right), \frac{\sigma^2}{t}\right]$$

Black-Scholes-Merton Equation

Assumptions:

- Frictionless markets: zero transaction cost (zero commission, zero taxation, zero bid-ask spread, zero market impact in trading), unlimited stock borrowing at zero cost, unlimited borrow and lending at risk free interest rate
- Stock price movement can be represented by a lognormal process:

  $dS = \mu S dt + \sigma S dW_t$

  - small increment in small time intervals → no "jumps"

  - symmetric distribution in stock return → no "fat tail"

As in the binomial tree case, we try to hedge our derivatives position with shares to form a riskless process:

Let f = derivatives instrument matures at T, S = share price, m = # of shares, r = continuous compounded risk free rate, div = 0,

$\Pi = f - mS$ (hedging f with m shares of S)

For $0 \le t < T$, under small time increment δt, $f \to f + \delta f$, $S \to S + \delta S$

$$\delta\Pi = (\delta f - m\delta S) = \left(\frac{\partial f}{\partial t} + \mu S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2}\right)\delta t + \sigma S\frac{\partial f}{\partial S}\delta W_t - m(\mu S\delta t + \sigma S\delta W_t)$$

$$\rightarrow \quad \delta\Pi = \left(\frac{\partial f}{\partial t} + \mu S\left(\frac{\partial f}{\partial S} - m\right) + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2}\right)\delta t + \sigma S\left(\frac{\partial f}{\partial S} - m\right)\delta W_t$$

If select m such that $m = \dfrac{\partial f}{\partial S}$, then the term contains δW drops out, the change of Π is thus no longer random:

$$\rightarrow \quad \delta\Pi = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2}\right)\delta t$$

Π becomes (instantaneously) riskless, hence its return rate must be r:

$$\rightarrow \quad \delta\Pi = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2}\right)\delta t = r\Pi\delta t = r\left(f - S\frac{\partial f}{\partial S}\right)\delta t$$

→ $\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} - rf = 0$ (Black-Scholes-Merton Equation)

Final condition:

- f = European call with strike $K = c$ → $c(S_T, T) = \max(S_T - K, 0)$
- f = European put with strike $K = p$ → $p(S_T, T) = \max(K - S_T, 0)$

Alternative derivation via replication:

Let g be a portfolio consists of m shares of S and an amount of cash $A = g - mS$. We want to use g to replicate f:

$$\delta g = m\left(\mu S\delta t + \sigma S\delta W_t\right) + r(g - mS)\delta t = rg\delta t + m(\mu - r)S\delta t + m\sigma S\delta W_t$$

Consider discounted g and f, $e^{-rt}g$ and $e^{-rt}f$:

$$\delta\left(e^{-rt}g\right) = -re^{-rt}g\delta t + e^{-rt}\delta g = e^{-rt}\left(m(\mu - r)S\delta t + m\sigma S\delta W_t\right) = m\delta\left(e^{-rt}S\right)$$

$$\delta\left(e^{-rt}f\right) = e^{-rt}\left(\frac{\partial f}{\partial t} + \mu S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2} - rf\right)\delta t + e^{-rt}\sigma S\frac{\partial f}{\partial S}\delta W_t$$

If $g = f$ at $t = 0$ and $\delta\left(e^{-rt}g\right) = \delta\left(e^{-rt}f\right)$ for all $0 < t < T$, → g replicates f

δW_t term: select m such that $m = \frac{\partial f}{\partial S}$

δt term: let $\frac{\partial f}{\partial t} + \mu S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2} - rf = m(\mu - r)S$

→ $\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2} - rf = 0$ (Black-Scholes-Merton Equation)

Since

$$\delta g = \delta f = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2}\right)\delta t + \frac{\partial f}{\partial S}\delta S$$

→ replication is "self financing"

Solution of BSM equation:

BSM equation is a parabolic PDE, similar to the diffusion equation in physics.

- European call:

$$c(S,t) = SN(d_1) - Ke^{-r(T-t)}N(d_2)$$

- European put:

$$p(S,t) = -SN(-d_1) + Ke^{-r(T-t)}N(-d_2)$$

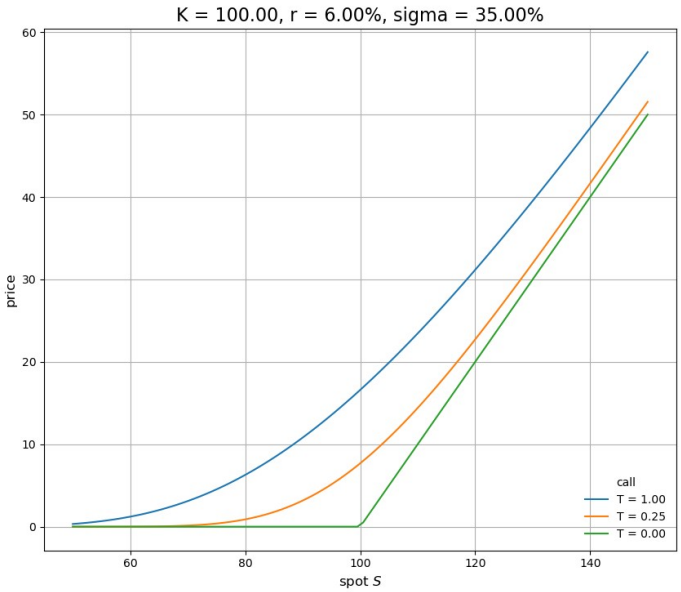
where $S \equiv S_t$, $N(x)$ is the CDF (cumulative distribution function) of $N(0,1)$,

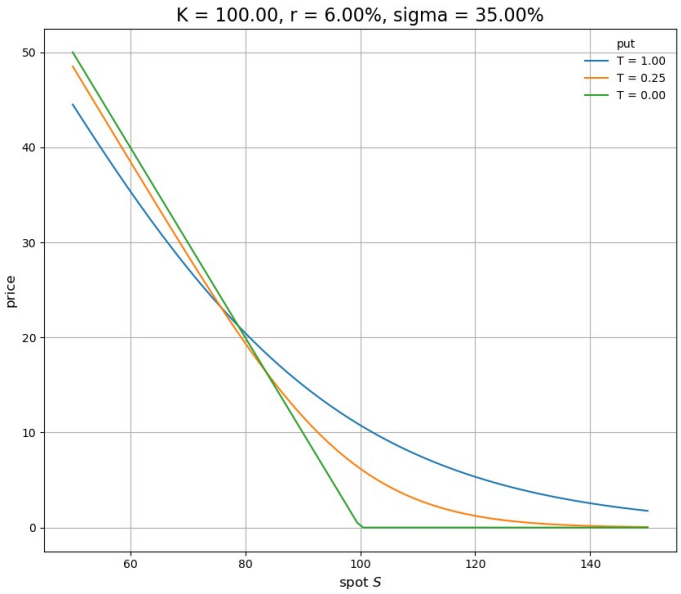
$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{u^2}{2}} du$$

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = \frac{\ln\left(\frac{S}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}} = d_1 - \sigma\sqrt{T-t}$$

Numerical examples:





Limit behavior:

- $S \to 0$: $(K > 0)$

  $d_1 \to -\infty, d_2 \to -\infty$ → $c \to 0, p \to Ke^{-r(T-t)}$

- $S \to \infty$:

  $d_1 \to \infty, d_2 \to \infty$ → $c \to S - Ke^{-r(T-t)}, p \to 0$

- $t \to T$:

  if $S > K$: $d_1 \to \infty, d_2 \to \infty$ → $c \to S - K$, $p \to 0$

  if $S < K$: $d_1 \to -\infty, d_2 \to -\infty$ → $c \to 0$, $p \to K - S$

  → $c \to \max(S - K, 0), p \to \max(K - S, 0)$

- $\sigma \to 0$

  if $\ln\left(\frac{S}{K}\right) + r(T-t) > 0 \Leftrightarrow S > Ke^{-r(T-t)}$: $d_1 \to \infty, d_2 \to \infty$

  → $c \to S - Ke^{-r(T-t)}, p \to 0$

if $\ln\left(\frac{S}{K}\right)+r(T-t)<0 \Leftrightarrow S<Ke^{-r(T-t)}$: $d_1\to-\infty, d_2\to-\infty$

→ $c\to 0$, $p\to Ke^{-r(T-t)}-S$

→ $c\to\max(S-Ke^{-r(T-t)},0), p\to\max(Ke^{-r(T-t)}-S,0)$

Put-call parity:

$N(-x)=1-N(x)$ → put-call parity is indeed satisfied by BSM formula

<u>BSM formula via risk neutral pricing</u>:

Stock drift μ does not appear in the BSM equation → risk neutral pricing

Consider a call:

$c(S,t)=e^{-r(T-t)}E_t\left[c(S_T,T)\right]=e^{-r(T-t)}E_t\left[\max(S_T-K,0)\right]$

$E_t[.]$ = expectation under risk neutral probability:

$$\ln S_T\sim N\left[\ln S_t+\left(r-\frac{\sigma^2}{2}\right)(T-t),\sigma^2(T-t)\right]$$

→ $$P(S_T\le Y)=\int_0^Y\frac{1}{\sigma x\sqrt{2\pi(T-t)}}e^{-\frac{\left(\ln x-\ln S_t-\left(r-\frac{\sigma^2}{2}\right)(T-t)\right)^2}{2\sigma^2(T-t)}}dx$$

Payoff at maturity

$$c(S_T,T)=\max(S_T-K,0)=S_TI_K-KI_K\ ,\ I_K=\begin{cases}1 & S_T>K\\ 0 & S_T\le K\end{cases}$$

→ $c(S,t)=e^{-r(T-t)}E_t\left[S_TI_K\right]-e^{-r(T-t)}KE_t\left[I_K\right]$

$e^{-r(T-t)}E_t\left[S_TI_K\right]$ = PV of expected stock received upon exercising call = $SN(d_1)$

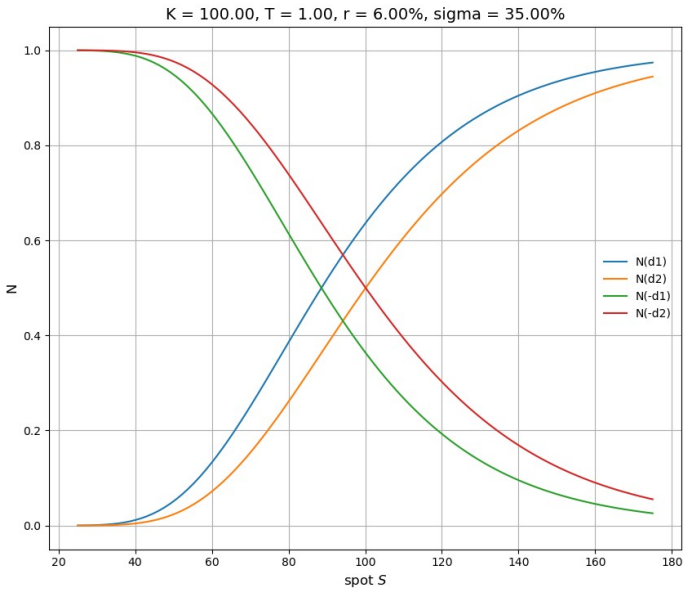
$e^{-r(T-t)}KE_t\left[I_K\right]$ = PV(K) × probability of call end up ITM = $Ke^{-r(T-t)}N(d_2)$

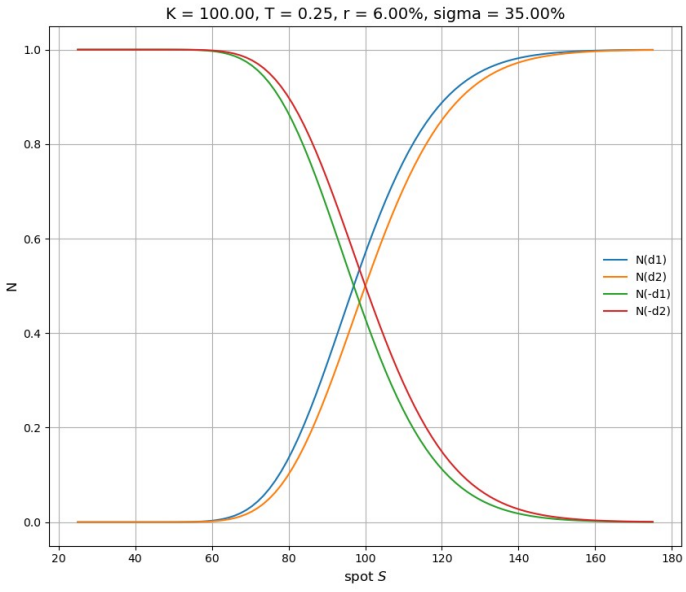
→ $c(S,t)=SN(d_1)-Ke^{-r(T-t)}N(d_2)$ (exercise)

Similarly for put

$$p(S,t) = -SN(-d_1) + Ke^{-r(T-t)}N(-d_2) \qquad \text{(exercise)}$$

Plot of $N(d_1), N(d_2), N(-d_1), N(-d_2)$ in the previous numerical example:





BSM formula as limit of Binomial Tree:

Let $\delta t = \frac{T-t}{N}$, construct a N-step binomial tree following the cookbook describe in the previous lecture:

$N \to \infty$ → $\delta t \to 0$ → BSM formula (Hull appendix of Chapter 13)

Summary:

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BSM option pricing formula may be arrived via

- solution of BSM partial differential equation which can be derived through
  - Constructing a (instantaneous) delta neutral portfolio
  - Dynamic replication of option payoff
- risk neutral pricing (more in IEOR 222)
- limiting case of binomial tree pricing

Exercise:

Verify that the BSM formulas are indeed solutions for the BSM differential equation subjects to the appropriate final conditions.

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For a derivative product f ($f = c, p$ discussed so far), BSM shows $f(K,T,t,S,r,\sigma)$,

- K,T ← contract
- S,r ← market quotes
- σ ← ?